

Technical Model Evidence

Survival Strategy Framework - Synthetic Retention Demo

Run conclusion: PASS_WITH_REVIEW

The Survival Strategy Framework - Synthetic Retention Demo run analyzed 7,500 records with an observed event rate of 39.52%. The regularized CoxPH model produced an apparent development-sample concordance index of 0.827; deterministic 5-fold validation produced a mean concordance of 0.827 (SD 0.007). The baseline high risk (top 25%) rule selected 1,875 records for same-cohort sensitivity testing. Among the illustrative improvement scenarios, 'Combined Retention Strategy' produced the largest modeled movement for the same 1,875-record target cohort: mean partial hazard declined by 52.86%, and modeled survival at 12 months increased by 18.79 percentage points. The run concluded PASS_WITH_REVIEW; proportional-hazards evidence was classified PASS_WITH_REVIEW. These outputs are model-based sensitivity evidence from demonstration data, not causal treatment effects, production retention forecasts, or realized financial outcomes.

1. Acceptance Checks

Check	Status	Evidence
Unique governed IDs	PASS	7,500 unique IDs across 7,500 rows.
Target cohort is non-empty	PASS	1,875 records selected at quantile 0.75.
Same target cohort across scenarios	PASS	Every scenario score table was compared with the identical baseline target IDs.
Engineered dependencies synchronized	PASS	All interaction and squared-term post-rebuild checks were reviewed.
Scenario change audit completed	PASS	Configured changes, realized movement, and bound-hit rates were archived.
Scenario metrics are calculated and finite	PASS	No hard-coded uncertainty, ROI, revenue, or defended-value metric is used.
Predicted survival remains within [0, 1]	PASS	All baseline and scenario survival probabilities satisfy probability bounds.
Predicted survival begins at S(0)=1	PASS	The first baseline and scenario survival values were tested at time zero.
Scenario direction matches configured expectation	PASS	Control, favorable, stress, and technical scenarios were compared with expected direction.
No-change control scenario	PASS	Expected zero modeled movement within numerical tolerance.
Adverse stress scenario	PASS	Expected negative hazard-reduction and survival-uplift evidence.
Cross-validation completed	PASS	5 stratified folds were required.
Out-of-fold calibration evidence	PASS	Maximum group absolute calibration error: 7.08%.
Persona quality evidence	PASS	Silhouette=0.161; mean stability ARI=0.992.
Proportional-hazards diagnostics	REVIEW	Global documented PH status: PASS_WITH_REVIEW.
PH sensitivity review	PASS	Low-cardinality raw PH review features were stratified where technically appropriate; target-cohort overlap and favorable-scenario ranking were compared.
Substantive reproducibility	PASS	Persona, model, validation, target-cohort, scenario, and curve outputs were repeated.
Executive and technical reports created	PASS	Survival_Strategy_Deck.pptx Technical_Model_Evidence.pdf

2. Input Validation

Severity	Area	Field	Message
INFO	Event	attrition_event	Observed event rate is 39.52%; right-censoring rate is 60.48%.

3. Model Fit Metadata

Engine_Release	Rows	Events	Censored	Event_Rate	Feature_Count	Penalizer	L1_Ratio	Standardized_Features	Apparent_Concordance_Index	CV_Folds	CV_Mean_Concordance	CV_SD_Concordance	CV_Min_Concordance	CV_Max_Concordance	CV_Status	Integrated_Brier_Score	Max_Group_Absolute_Calibration_Error	Calibration_Status	PH_Global_Status	Partial_AIC	Log_Likelihood	Fit_Status
2026.07-mainline	7500	2964	4536	39.52%	12	0.1000	0.0000	True	0.8272	5	0.8272	0.0069	0.8196	0.8364	PASS	11.88%	7.08%	PASS	PASS_WITH_REVIEW	46,523.7018	-23,249.8509	SUCCESS

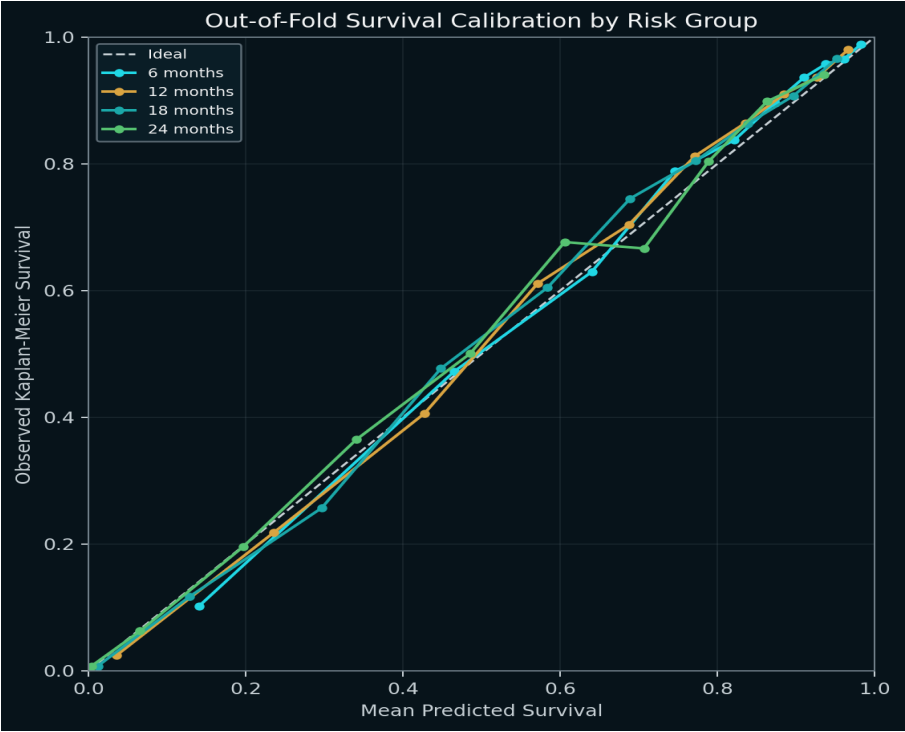
4. Stratified K-Fold Concordance

Fold	Train_Rows	Validation_Rows	Validation_Events	Validation_Event_Rate	Concordance_Index	Status	Error
1	6000	1500	592	39.47%	0.8364	PASS	
2	6000	1500	593	39.53%	0.8218	PASS	
3	6000	1500	593	39.53%	0.8196	PASS	
4	6000	1500	593	39.53%	0.8266	PASS	
5	6000	1500	593	39.53%	0.8314	PASS	

5. Out-of-Fold Calibration Metrics

Horizon	Time_Unit	At_Risk_At_Horizon	Observed_Events_By_Horizon	Censored_By_Horizon	Overall_Predicted_Survival	Overall_Observed_KM_Survival	Calibration_Difference	Absolute_Calibration_Error	IPCW_Brier_Score	Brier_Score_Method	Integrated_Brier_Score	Integrated_Brier_Score_Scope	Integration_Horizon_Min	Integration_Horizon_Max	Max_Group_Absolute_Calibration_Error	Calibration_Review_Threshold	Calibration_Status
6.0000	months	5678	1822	0	74.76%	75.71%	-0.95%	0.95%	0.1102	IPCW	11.88%	Trapezoidal average across configured horizons; not a full continuous-time IBS.	6.0000	24.0000	7.08%	0.1000	PASS
12.0000	months	3603	2588	1309	63.40%	64.63%	-1.23%	1.23%	0.1302	IPCW	11.88%	Trapezoidal average across configured horizons; not a full continuous-time IBS.	6.0000	24.0000	7.08%	0.1000	PASS
18.0000	months	1550	2884	3066	56.16%	57.55%	-1.39%	1.39%	0.1332	IPCW	11.88%	Trapezoidal average across configured horizons; not a full continuous-time IBS.	6.0000	24.0000	7.08%	0.1000	PASS

Horizon	Time_Unit	At_Risk_At_Horizon	Observed_Events_By_Horizon	Censored_By_Horizon	Overall_Predicted_Survival	Overall_Observed_KM_Survival	Calibration_Difference	Absolute_Calibration_Error	IPCW_Brier_Score	Brier_Score_Method	Integrated_Brier_Score	Integrated_Brier_Score_Scope	Integration_Horizon_Min	Integration_Horizon_Max	Max_Group_Absolute_Calibration_Error	Calibration_Review_Threshold	Calibration_Status
24.0000	months	0	2964	4536	49.90%	51.25%	-1.35%	1.35%	0.0757	IPCW	11.88%	Trapezoidal average across configured horizons; not a full continuous-time IBS.	6.0000	24.0000	7.08%	0.1000	PASS



6. CoxPH Model Summary

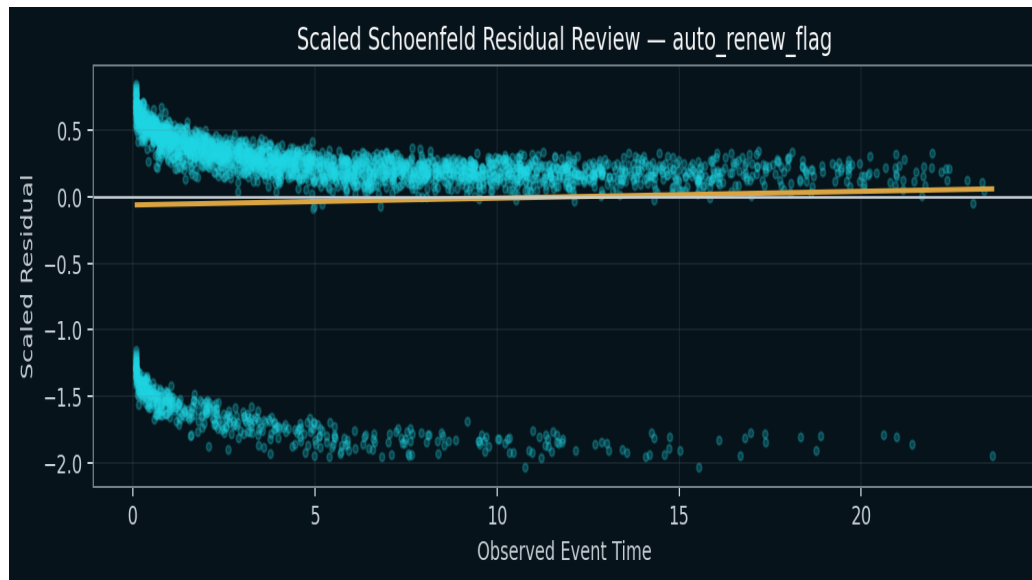
Feature	Coefficient	Hazard_Ratio	Hazard_Ratio_CI_Lower	Hazard_Ratio_CI_Upper	P_Value	Interpretation_Unit
onboarding_completion	-0.3641	0.6948	0.6595	0.7320	1.168e-42	Per 1 SD
monthly_active_days	-0.2966	0.7433	0.7136	0.7743	4.523e-46	Per 1 SD
product_adoption_count	-0.1417	0.8679	0.8278	0.9099	4.315e-09	Per 1 SD
support_tickets_90d	0.1422	1.1528	1.1042	1.2036	9.788e-11	Per 1 SD
support_resolution_days	0.2292	1.2575	1.2097	1.3073	5.249e-31	Per 1 SD
service_incidents_90d	0.1772	1.1938	1.1574	1.2314	4.245e-29	Per 1 SD
tenure_at_start	-0.1109	0.8951	0.8658	0.9253	5.822e-11	Per 1 SD
auto_renew_flag	-0.2363	0.7895	0.7681	0.8115	1.083e-63	Per 1 SD
enterprise_flag	-0.1120	0.8940	0.8622	0.9270	1.301e-09	Per 1 SD
monthly_active_days_x_product_adoption_count	-0.2077	0.8125	0.7672	0.8604	1.274e-12	Per 1 SD
support_tickets_90d_x_support_resolution_days	0.2109	1.2348	1.1798	1.2923	1.071e-19	Per 1 SD
onboarding_completion_sq	-0.3109	0.7328	0.6931	0.7748	8.135e-28	Per 1 SD

7. Proportional-Hazards Tests

Feature	Test_Statistic	P_Value	P_Value_Holm	P_Value_BH	Raw_PH_Status	Holm_PH_Status	PH_Assumption_Status	Review_Rationale
auto_renew_flag	4.8431	0.02776	0.3331	0.3331	REVIEW	PASS	PASS_WITH_RAW_REVIEW	Raw p-value is below alpha but multiplicity-adjusted tests pass; retain as documented review item.
enterprise_flag	0.3292	0.5661	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
monthly_active_days	0.1355	0.7128	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
monthly_active_days_x_product_adoption_count	0.0045	0.9464	1	0.9464	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
onboarding_completion	0.7814	0.3767	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
onboarding_completion_sq	0.0764	0.7823	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
product_adoption_count	0.5026	0.4784	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
service_incidents_90d	0.1471	0.7013	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
support_resolution_days	2.7018	0.1002	1	0.6014	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
support_tickets_90d	0.2117	0.6454	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
support_tickets_90d_x_support_resolution_days	1.1336	0.287	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.
tenure_at_start	0.7514	0.386	1	0.8534	PASS	PASS	PASS	No feature-level PH review item at the configured alpha.

8. PH Sensitivity Review

Feature	Raw_P_Value	Holm_Adjusted_P_Value	Holm_PH_Status	Feature_Uniq_Levels	Sensitivity_Method	Fit_Status	Apparent_Concordance_Index	Concordance_Delta	Target_Cohort_Overlap_Rate	Target_Cohort_Jaccard	Scenario_Rank_Spearman	Scenario_Rank_Exact_Match	Primary_Best_Improvement	Sensitivity_Best_Improvement	Scenario_Rank_Status	Conclusion
auto_renew_flag	0.02776	0.3331	PASS	2	STRATIFIED_COXPH	PASS	0.8167	-0.0105	93.71%	0.8816	1.0000	True	Combined Retention Strategy	Combined Retention Strategy	PASS	Automatic stratified sensitivity fit completed. Review discrimination, target-cohort overlap, and improvement-scenario ranking before changing the primary specification.

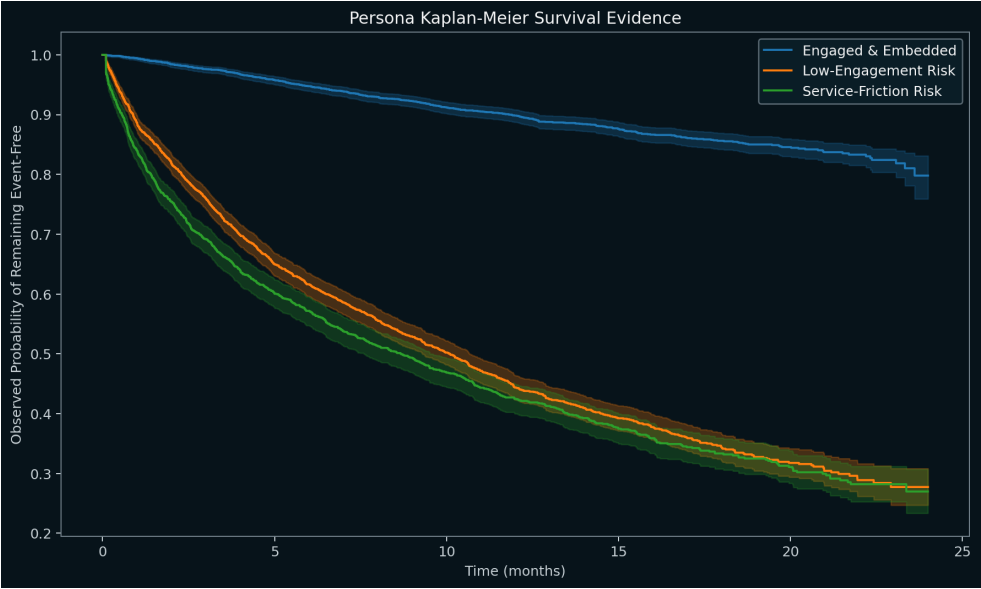


9. Persona Quality and Stability

Rows	Clusters	Inertia	Silhouette_Score	Silhouette_Sample_Size	Davies_Bouldin_Index	Calinski_Harabasz_Index	Smallest_Cluster_Rate	Largest_Cluster_Rate	Stability_Mean_AR	Stability_Min_AR	Silhouette_Review_Threshold	Persona_Quality_Status
7500	3	37,848.5652	0.1608	2000	1.8752	1,451.0763	21.99%	45.45%	0.9920	0.9799	0.1000	PASS
Repeat				Random_State					Adjusted_Rand_Index_vs_Baseline			
1				43					0.9978			
2				44					0.9983			
3				45					0.9996			
4				46					0.9843			
5				47					0.9799			

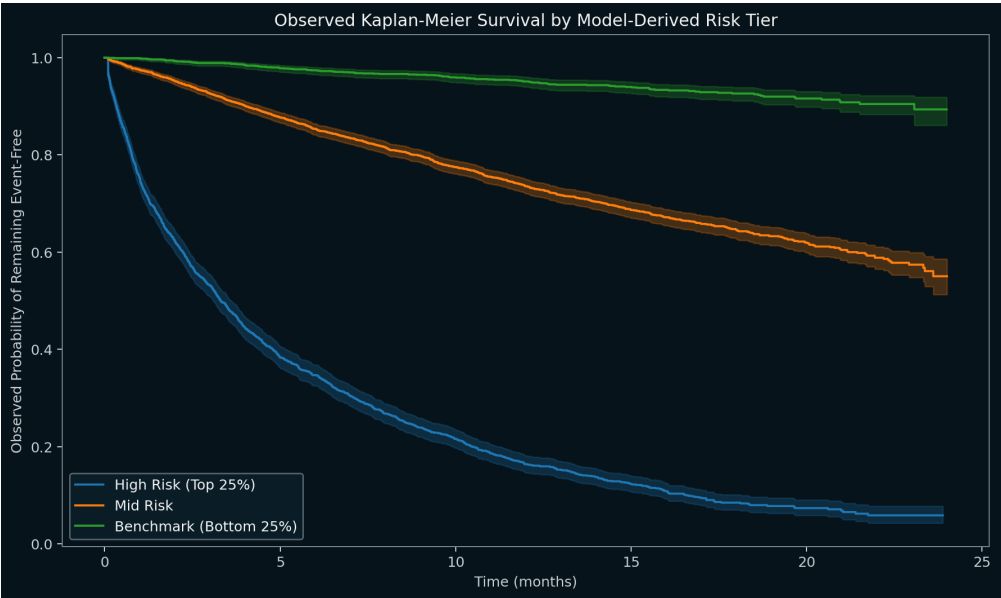
10. Persona Profiles

Segment_ID	Persona	Record_Count	Event_Rate	Median_Duration	onboarding_completion	monthly_active_days	product_adoption_count	support_tickets_90d	support_resolution_days	service_incidents_90d	tenure_at_start	auto_renew_flag
0	Service-Friction Risk	1649	63.19%	8.3088	0.6386	16.4461	3.5864	2.9739	4.4174	1.5367	13.0537	0.9030
1	Engaged & Embedded	3409	12.94%	14.7651	0.8069	20.2119	4.6480	0.9223	2.3468	0.3878	13.3197	0.9451
2	Low-Engagement Risk	2442	60.65%	8.9214	0.5438	13.5778	2.6744	1.3542	2.5816	0.3825	12.3153	0.8657



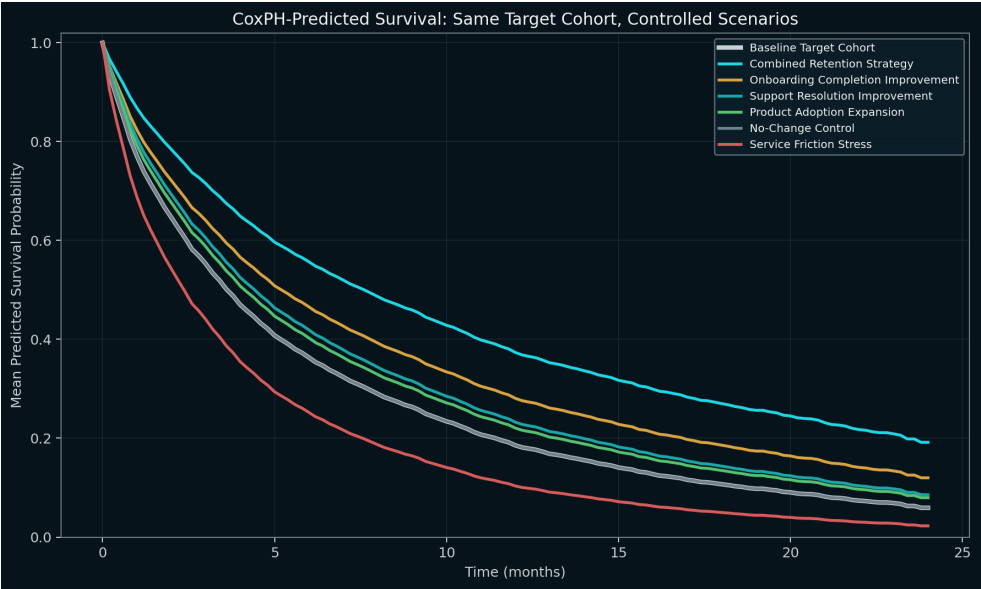
11. Risk-Tier Profiles

Risk_Tier	Record_Count	Event_Rate	Median_Duration	Mean_Partial_Hazard	onboarding_completion	monthly_active_days	product_adoption_count	support_tickets_90d	support_resolution_days	service_incidents_90d	tenure_at_start	auto_renew_flag
High Risk (Top 25%)	1875	87.57%	3.3057	10.8997	0.4909	13.0865	2.6379	2.4363	3.8778	0.9579	11.3277	0.7680
Mid Risk	3750	31.92%	12.7099	1.0978	0.6927	17.2074	3.6491	1.4133	2.8128	0.6131	13.0031	0.9467
Benchmark (Bottom 25%)	1875	6.67%	15.2466	0.2003	0.8605	21.3942	5.1520	0.7931	2.0104	0.3707	14.4029	0.9787



12. Scenario Results

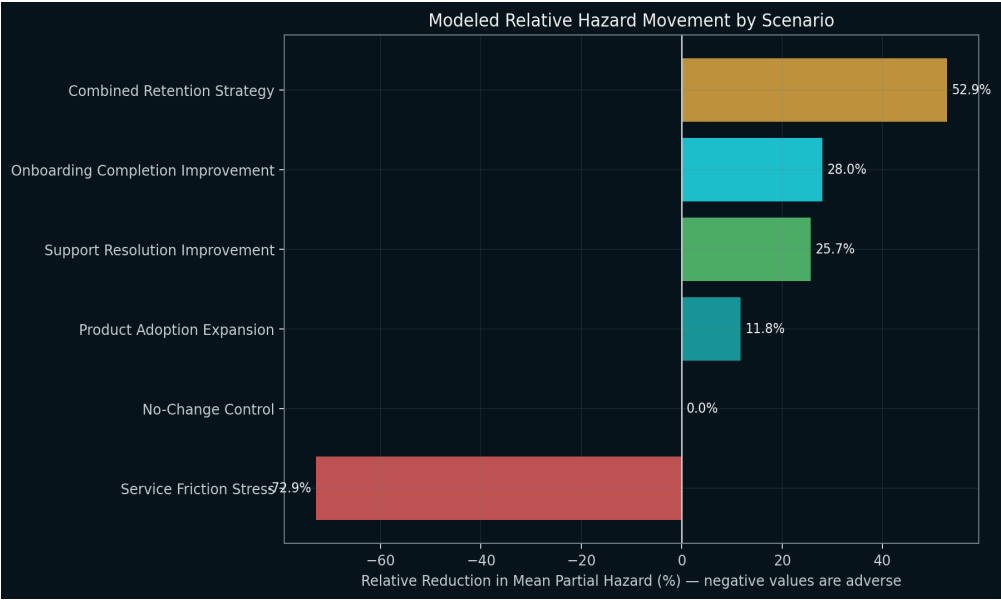
Scenario_Rank	Improvement_Rank	Scenario_Name	Scenario_Type	Target_Cohort_Count	Relative_Hazard_Reduction	Baseline_Survival_At_Horizon	Scenario_Survival_At_Horizon	Survival_Uplift_Percentage_Points	Modeled_Direction	Direction_Check_Status
1	1.0000	Combined Retention Strategy	IMPROVEMENT	1875	52.86%	18.56%	37.35%	18.79 pp	IMPROVED	PASS
2	2.0000	Onboarding Completion Improvement	IMPROVEMENT	1875	28.03%	18.56%	28.07%	9.51 pp	IMPROVED	PASS
3	3.0000	Support Resolution Improvement	IMPROVEMENT	1875	25.72%	18.56%	23.25%	4.69 pp	IMPROVED	PASS
4	4.0000	Product Adoption Expansion	IMPROVEMENT	1875	11.77%	18.56%	22.06%	3.49 pp	IMPROVED	PASS
5		No-Change Control	CONTROL	1875	0.00%	18.56%	18.56%	0.00 pp	NEUTRAL	PASS
6		Service Friction Stress	STRESS	1875	-72.86%	18.56%	10.34%	-8.22 pp	ADVERSE	PASS



13. Scenario Change Audit

Scenario_	Scenario_	Expected	Feature	Operation	Configure	Lower_Bo	Upper_Bo	Round_Di	Target_C	Records_	Records_	Records_	Lower_Bo	Records_	Upper_Bo	Mean_Ba	Mean_Un	Mean_Sc	Mean_Ab	Mean_Sig	Mean_Rel	Configure
Name	Type	_Direction			d_Value	und	und	gits	ohort_Cou	Changed	Changed_	Hitting_Lo	und_Hit_	Hitting_Up	und_Hit_	seline_Val	constraine	enario_Va	solute_Ch	ned_Chan	ative_Cha	d_vs_Rea
									nt		Rate	wer_Boun	Rate	per_Boun	Rate	ue	d_Value	lue	ange	ge	nge	lized_Not
												d		d								e
No-Chang e Control	CONTRO L	NEUTRAL	NO_CHA NGE	NONE	0.0000				1875	0	0.00%	0	0.00%	0	0.00%				0.0000	0.0000	0.00%	No-chang e control; no feature movement configured .
Onboardin g Complet ion Improv ement	IMPROVE MENT	IMPROVE D	onboardin g_complet ion	add	0.1000	0.0000	1.0000	6.0000	1875	1875	100.00%	0	0.00%	3	0.16%	0.4909	0.5909	0.5909	0.1000	0.1000	22.47%	Bounds and rounding may cause realized movement to differ from the configured change.
Support R esolution I mprovement	IMPROVE MENT	IMPROVE D	support_r esolution_ days	multiply	0.7500	0.2500	20.0000	4.0000	1875	1868	99.63%	8	0.43%	0	0.00%	3.8778	2.9084	2.9086	0.9692	-0.9692	-24.91%	Bounds and rounding may cause realized movement to differ from the configured change.
Product Adoption Expansion	IMPROVE MENT	IMPROVE D	product_a doption_ count	add	1.0000	1.0000	8.0000	0.0000	1875	1866	99.52%	0	0.00%	40	2.13%	2.6379	3.6379	3.6331	0.9952	0.9952	53.09%	Bounds and rounding may cause realized movement to differ from the configured change.
Combined Retention Strategy	IMPROVE MENT	IMPROVE D	onboardin g_complet ion	add	0.1000	0.0000	1.0000	6.0000	1875	1875	100.00%	0	0.00%	3	0.16%	0.4909	0.5909	0.5909	0.1000	0.1000	22.47%	Bounds and rounding may cause realized movement to differ from the configured change.
Combined Retention Strategy	IMPROVE MENT	IMPROVE D	support_r esolution_ days	multiply	0.7500	0.2500	20.0000	4.0000	1875	1868	99.63%	8	0.43%	0	0.00%	3.8778	2.9084	2.9086	0.9692	-0.9692	-24.91%	Bounds and rounding may cause realized movement to differ from the configured change.

Scenario_ Name	Scenario_ Type	Expected _Direction	Feature	Operation	Configure d_Value	Lower_Bo und	Upper_Bo und	Round_Di gits	Target_C ohort_Cou nt	Records_ Changed	Records_ Changed_ Rate	Records_ Hitting_Lo wer_Boun d	Lower_Bo und_Hit Rate	Records_ Hitting_Up per_Boun d	Upper_Bo und_Hit_ Rate	Mean_Ba seline_Val ue	Mean_Un constraine d_Value	Mean_Sc enario_Va lue	Mean_Ab solute_Ch ange	Mean_Sig ned_Chan ge	Mean_Rel ative_Ch ange	Configure d_vs_Rea lized_Not e
Combined Retention Strategy	IMPROVE MENT	IMPROVE D	product_a doption_c ount	add	1.0000	1.0000	8.0000	0.0000	1875	1866	99.52%	0	0.00%	40	2.13%	2.6379	3.6379	3.6331	0.9952	0.9952	53.09%	Bounds and rounding may cause realized movement to differ from the configured change.
Service Friction Stress	STRESS	ADVERS E	support_r esolution_ days	multiply	1.2500	0.2500	20.0000	4.0000	1875	1875	100.00%	0	0.00%	0	0.00%	3.8778	4.8473	4.8473	0.9695	0.9695	25.00%	Bounds and rounding may cause realized movement to differ from the configured change.
Service Friction Stress	STRESS	ADVERS E	service_in cidents_9 0d	add	1.0000	0.0000	8.0000	0.0000	1875	1875	100.00%	0	0.00%	0	0.00%	0.9579	1.9579	1.9579	1.0000	1.0000	77.94%	Bounds and rounding may cause realized movement to differ from the configured change.



14. Dependency Synchronization Audit

Scenario_Name	Engineered_Feature	Dependency_Type	Source_Features	Pre_Rebuild_Max_Abs_Diff	Post_Rebuild_Max_Abs_Diff	Recalculated	Validation_Status
No-Change Control	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	0.0000	0.0000	True	PASS
No-Change Control	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	0.0000	0.0000	True	PASS
No-Change Control	onboarding_completion_sq	SQUARE	onboarding_completion	0.0000	0.0000	True	PASS
Onboarding Completion Improvement	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	0.0000	0.0000	True	PASS
Onboarding Completion Improvement	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	0.0000	0.0000	True	PASS
Onboarding Completion Improvement	onboarding_completion_sq	SQUARE	onboarding_completion	0.1871	0.0000	True	PASS
Support Resolution Improvement	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	0.0000	0.0000	True	PASS
Support Resolution Improvement	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	16.7685	0.0000	True	PASS
Support Resolution Improvement	onboarding_completion_sq	SQUARE	onboarding_completion	0.0000	0.0000	True	PASS
Product Adoption Expansion	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	25.3890	0.0000	True	PASS
Product Adoption Expansion	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	0.0000	0.0000	True	PASS
Product Adoption Expansion	onboarding_completion_sq	SQUARE	onboarding_completion	0.0000	0.0000	True	PASS
Combined Retention Strategy	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	25.3890	0.0000	True	PASS
Combined Retention Strategy	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	16.7685	0.0000	True	PASS
Combined Retention Strategy	onboarding_completion_sq	SQUARE	onboarding_completion	0.1871	0.0000	True	PASS
Service Friction Stress	monthly_active_days_x_product_adoption_count	INTERACTION	monthly_active_days product_adoption_count	0.0000	0.0000	True	PASS
Service Friction Stress	support_tickets_90d_x_support_resolution_days	INTERACTION	support_tickets_90d support_resolution_days	16.7685	0.0000	True	PASS
Service Friction Stress	onboarding_completion_sq	SQUARE	onboarding_completion	0.0000	0.0000	True	PASS

15. Reproducibility Checks

Check	Status	Evidence
Persona assignments reproducible	PASS	Segment_ID assignments were compared row-for-row under the same seed and configuration.
CoxPH coefficients reproducible	PASS	Maximum absolute coefficient difference: 0.000e+00.
Cross-validation evidence reproducible	PASS	Maximum absolute fold concordance difference: 0.000e+00.
Calibration evidence reproducible	PASS	Maximum absolute calibration difference: 0.000e+00.
PH diagnostic evidence reproducible	PASS	Maximum absolute raw/adjusted PH p-value difference: 0.000e+00; status match=True.
Target cohort reproducible	PASS	Compared 1,875 baseline target IDs with the repeated run.
Scenario evidence reproducible	PASS	Maximum absolute scenario metric difference: 0.000e+00; directions match=True.

Check	Status	Evidence
Predicted survival curves reproducible	PASS	Maximum absolute curve difference: 0.000e+00.
PH sensitivity evidence reproducible	PASS	Maximum absolute PH sensitivity metric difference: 0.000e+00; categorical evidence match=True.

16. Interpretation Boundaries

- The analysis uses synthetic or public demonstration data.
- Scenario movement is modeled sensitivity, not causal treatment effect.
- Partial hazard is relative risk, not an event probability.
- Apparent concordance is in-sample; cross-validation is reported separately.
- Out-of-fold calibration does not replace external or temporal validation.
- Predicted survival requires domain validation before production use.
- No realized revenue, ROI, or defended LTV is claimed.